

9.3 Latency & throughput of atomic operations

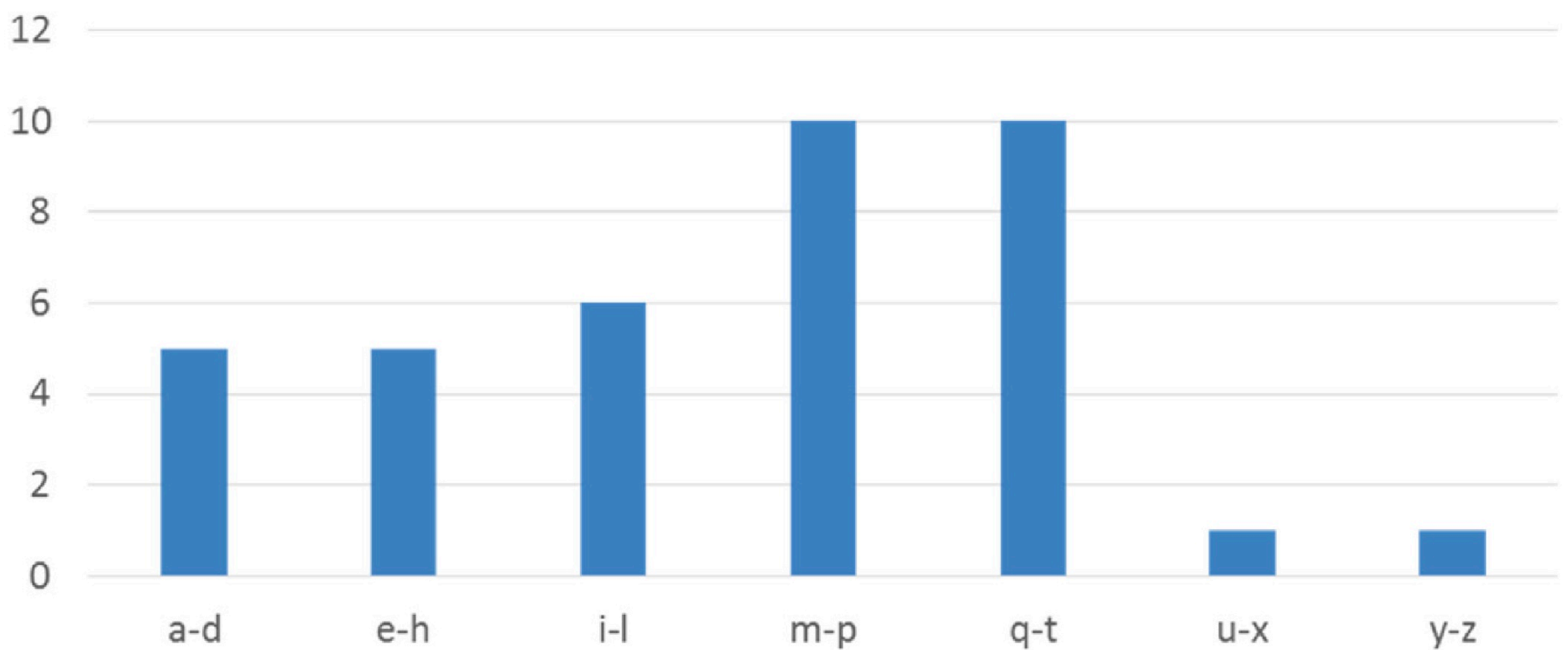
The Formula:

The theoretical effective throughput when dealing with contended atomic operations is calculated as:

$$\text{Effective Throughput} = \left(\frac{\text{Total Operations}}{\text{Max Contention in a Single Bin}} \right) \times \text{Base Throughput}$$

↑
Total no. of atomic operations

↓
Max No. of atomic ops that hit the same location

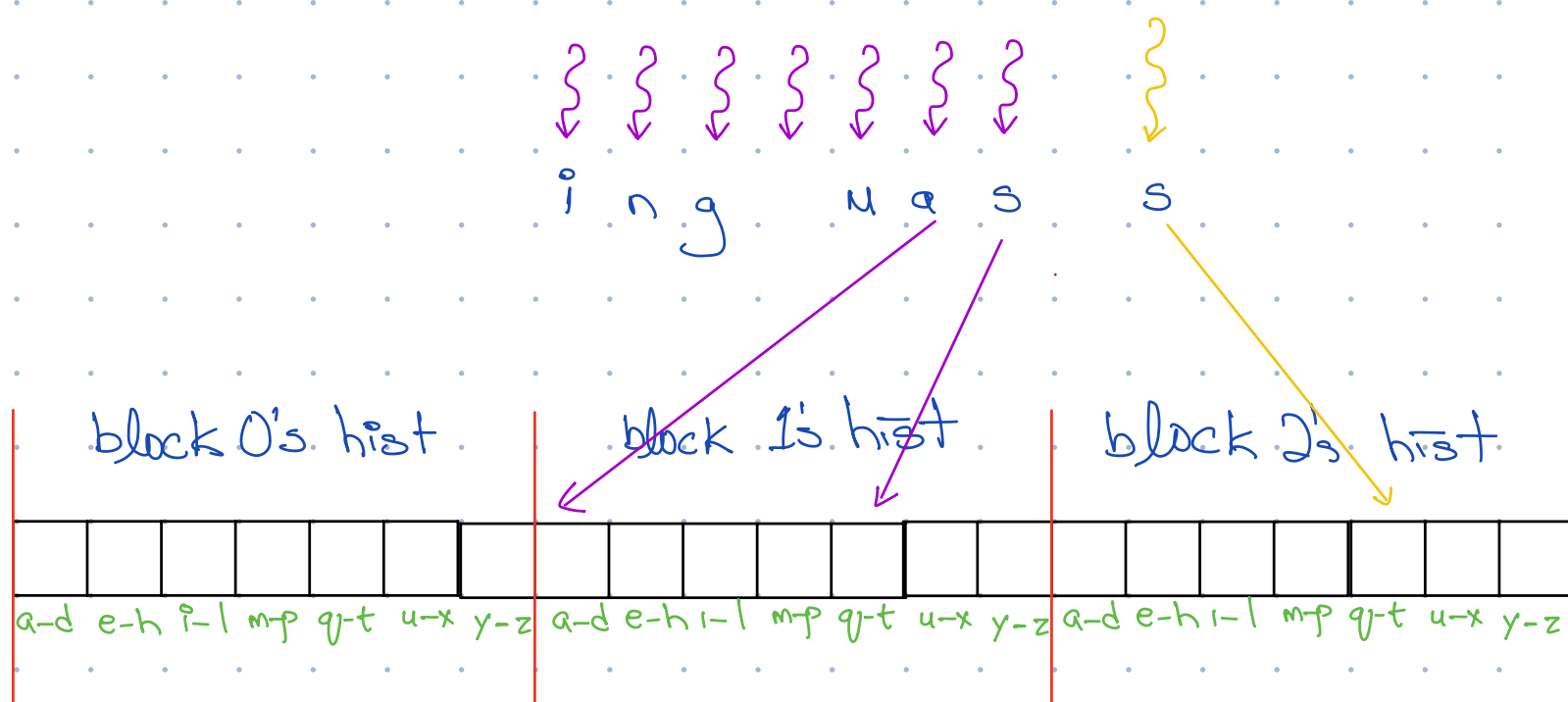


$$\text{Effective throughput} = \frac{5 + 5 + 6 + 10 + 10 + 1 + 1}{10} \times 2.5$$

$$= \frac{38}{10} \times 2.5$$

$$= 9.5 \text{M atoms/sec}$$

9.4 Privatization



$\text{hist}[\text{blockIdx} \cdot x * \text{NumBins} + \text{alphabet_pos}/4]++$

if ($\text{blockIdx} \cdot x > 0$)

}

for ($\text{bin} = \text{threadIdx} \cdot x$; $\text{bin} < \text{NumBins}$; $\text{bin} += \text{blockDim} \cdot x$)

{

$\text{binVal} = \text{histo}[\text{blockIdx} \cdot x * \text{NumBins} + \text{bin}]$

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↳ All threads take their value they are responsible for & adds them to the hist of block 0's hist

↳ The $\text{bin} += \text{blockDim} \cdot x$ handles the case when NumBins larger than block size.

NOTE:

- Threads are launched according to input data.
- Each thread is responsible for atmost one char (lines 2-8)
- Each thread is responsible for 0, 1, multiple bins (lines 9-18)

9.5 Thread Coarsening

$$\text{blockDim} * \text{blockIdx} + \text{threadIdx} * \text{segment}$$

$$100 \times 10^3 \text{ sec} \times \frac{1 \text{ nClock}}{\text{sec}}$$